

YOGESH GAUTAM, PE

Civil Engineer II, PE | Water Resources & Wastewater Modeling Specialist

Minneapolis, MN | Licensed Professional Engineer (PE), State of Minnesota | American Water Works Association Member

PROFESSIONAL SUMMARY

Professional Engineer with a Master's degree in Civil Engineering (Water Resources) and ten years of professional experience specializing in the planning, design, and management of municipal water, wastewater, and stormwater projects. Combines municipal public-works experience with consulting experience, spanning conveyance design, hydraulic modeling, pump and lift station design, water storage and distribution design, and stormwater management. Professional interests also include the application of advanced data analytics to integrate and process large-scale datasets from the water and wastewater industry, supporting real-time system monitoring and enhancing operational efficiency. Authored several publications in national and international journals, covering a range of topics including water infrastructure, wastewater treatment, water treatment and water quality modeling.

EDUCATION

M.S., Civil Engineering 2023
University of Kentucky, Lexington, KY, USA — GPA: 3.625

B.E., Civil Engineering 2016
Tribhuvan University, Nepal — 71.58%

LICENSURE & PROFESSIONAL MEMBERSHIPS

- Licensed Professional Engineer (PE), Civil Engineering — State of Minnesota
- Engineer-in-Training (EIT), National Council of Examiners for Engineering & Surveying—Commonwealth of Kentucky
- Member, American Water Works Association (AWWA)
- Member, American Society of Civil Engineers (ASCE)
- Member, Nepal Engineers' Association
- President, AWWA Water Professionals Student Chapter, University of Kentucky (Feb 2023 – May 2023)

PROFESSIONAL EXPERIENCE

Civil Engineer II, PE May 2025 – Present
Public Works, City of Saint Paul, MN

- Prepare and review stormwater and sanitary sewer designs, cost estimates, reports, applications, specifications, and plans for the City's street projects.
- Oversee analyses, computations, and design plans for hydraulic and drainage systems on transportation projects.
- Serve as Sewer Utility representative and Technical Advisory Committee (TAC) member for sewer design and construction on City, County, State, and private development projects.
- Perform hydraulic modeling of wastewater and stormwater collection systems across areas of the City of Saint Paul.
- Develop and refine stormwater management solutions, including detention/retention basin design, erosion control design, and Stormwater Pollution Prevention Plans (SWPPPs).
- Provide construction support for City wastewater collection system, lift station, and storm infrastructure projects.

Water Resources Engineer — Civil EIT May 2023 – Mar 2025
Stantec Inc., Columbus, OH

- Performed hydraulic modeling of municipal wastewater collection and conveyance systems using SewerGEMS, InfoWorks ICM, the EPA SSOAP Toolbox, and other software for utilities across the U.S.
- Performed hydraulic modeling and planning of municipal water distribution systems using WaterGEMS, InfoWater Pro, HAMMER, and other software for multiple U.S. cities.
- Revised hydraulic models and performed pump-sizing analyses for pump stations in Lexington, KY, including the Kentucky American Water Cox Street Booster Station and Ford-Hampton Booster Station; ran scenario analyses across demand and flow conditions to confirm regulatory compliance.
- Performed hydraulic analysis, modeling, and report preparation for sizing and elevation of water storage tanks in Tennessee, including the Bordeaux Hills Tank and Aetna Mountain Tank.

- Conducted hydrologic and hydraulic modeling of flow volume and water-quality performance for stormwater systems in new and redevelopment projects, including the Santee Cooper Winyah and Cross Generating Stations in South Carolina.
- Led hydraulic design of a new 15-MGD variable-frequency-drive pump for a 45-MGD raw water intake station on the Kentucky River (KRS1).
- Supported modeling and planning of a 12-mile watermain extension connecting satellite communities to the Lexington, KY system.

Civil Sub Engineer

Feb 2018 – Aug 2021

Water Resource and Irrigation Department, Government of Nepal

- Performed analysis, design, report preparation, costing, and documentation for medium- and small-scale irrigation projects using open channels and pipe conduits in the Jhapa and Taplejung districts.
- Provided construction support and site supervision for major irrigation structures, including headworks and off-taking canals, for rural irrigation projects.
- Designed and oversaw construction of river training structures and flood-control protection works for localized flooding in Jhapa, Nepal.
- Delivered water distribution and sprinkler irrigation projects supporting sustainable farming for rural communities in Eastern Nepal.

Civil Site Engineer

Apr 2016 – Jan 2018

District Level Planning Implementation Unit, Government of Nepal

- Supported post-earthquake reconstruction for rural communities in central Nepal following the 2015 earthquake.
- Conducted rapid visual inspections and surveys to assess structural damage and recommend reconstruction relief, Kathmandu, Nepal.
- Applied technical guidelines and standards for earthquake-resistant construction in rural communities, Kavrepalanchowk, Nepal.
- Worked directly with affected communities to understand needs and incorporate them into the reconstruction process, Kavrepalanchowk, Nepal.

SELECTED PROJECT EXPERIENCE

I. Current Projects

A. Hydraulic Modeling

Citywide Sanitary Sewer Modeling

City of Saint Paul, MN | City Modeling Lead

Development and maintenance of the citywide sanitary sewer hydraulic model, supporting capacity assessment, capital improvement planning, and evaluation of proposed development and street reconstruction projects across the sewer collection system. Cities usually rely on static modeling for assessment of their projects. However, my approach is developing hydrodynamic modeling to ensure that the varying hydraulic behavior of complex systems is accurately represented, enabling optimized infrastructure design, risk mitigation, and sustainable asset management.

B. Sanitary and Stormwater Utilities

In my current role as a Civil Engineer for the City of Saint Paul, Capital city of Minnesota, I design and oversee infrastructure projects that typically last 2 to 3 years on average to complete. In addition to design experience, I have performed infrastructure research, inspections and assisted in the development of evaluation reports to prevent infrastructure failure and engineered solutions to improve their serviceability and useful life. I integrate IWRM approach to incorporate sanitary and stormwater infrastructure rehabilitation with other infrastructure projects to reduce costs. I have also encouraged trenchless rehabilitation techniques like cured-in-place pipe (CIPP) lining, pipe bursting, spray-applied linings where possible to avoid costly traditional open-cut restoration methods.

Earl Street Reconstruction Phase 1 (Maryland Ave to Burns Ave)

City of Saint Paul, MN | 2025-2028 | Sewer Utility Representative

Represented the Sewer Utility in design review and utility coordination for sanitary and storm sewer improvements associated with street reconstruction. Trenchless rehabilitation techniques like cured-in-place pipe (CIPP) lining, spray-applied linings to avoid costly traditional open-cut restoration methods. Design and construction of stormwater volume reduction Best Management Practices (BMPs) treats millions of gallons of rainwater, keeping tons of sediment and harmful chemicals out of water bodies.

Robert Street Reconstruction (Annapolis St to Kellogg Blvd)

MnDOT | 2025–2027 | Sewer Utility Representative, TAC

Represented the Sewer Utility on the technical advisory committee for this MnDOT-led corridor reconstruction, reviewing and coordinating sanitary and storm sewer design and utility relocations along Robert Street. Incorporate sanitary and stormwater infrastructure rehabilitation with street reconstruction projects to reduce costs and improve infrastructure resiliency.

Jackson Street Reconstruction (University Ave to Pennsylvania Ave)

City of Saint Paul, MN | 2025 | Sewer Utility Representative, Construction Phase

Reviewed and coordinated sanitary and storm sewer design as part of full street reconstruction, representing the Sewer Utility during the project's construction phase. Provided technical guidance for constructing cost effective and practical stormwater management strategy.

University Avenue Reconstruction (I-35E to Lafayette Rd)

City of Saint Paul, MN | 2025–2026 | Sewer Utility Design and Construction

Provided sewer utility design review and construction support for sanitary and storm infrastructure along this major arterial reconstruction corridor. Provided technical guidance for developing cost effective and practical stormwater management strategy.

Pelham Boulevard Reconstruction (Mississippi River Blvd to Franklin Ave)

City of Saint Paul, MN | 2026 | Sewer Utility Representative

Provided sanitary and storm sewer design review and utility coordination as Sewer Utility representative on the project design and construction. Trenchless rehabilitation techniques like cured-in-place pipe (CIPP) lining, spray-applied linings to avoid tree harming & costly traditional open-cut restoration methods. Design and construction of stormwater volume reduction Best Management Practices (BMPs) treats millions of gallons of rainwater, keeping tons of sediment and harmful chemicals out of water bodies.

Arlington Avenue Reconstruction (Wheelock Pkwy to Payne Ave)

City of Saint Paul, MN | 2026–2028 | Sewer Utility Representative

Represented the Sewer Utility in reviewing sanitary and storm sewer design and coordinating sanitary and storm improvements along this corridor. Designing and reviewing stormwater volume reduction Best Management Practices (BMPs) treats millions of gallons of rainwater, keeping tons of sediment and harmful chemicals out of downstream water sources.

Summit Avenue Reconstruction (Mississippi River Blvd to Hamline Ave)

City of Saint Paul, MN | 2026–2028 | ~\$42.2M | Sewer Utility Design

Providing sanitary and storm sewer planning and design review and utility coordination for the full historic-corridor reconstruction.

C. Stormwater and Flood Protection

I have been assisting City of Saint Paul (City) meet stormwater management goals and refine current design and maintenance practices. I also provide technical reviews for developing cost effective and practical operation and maintenance strategies. I have been an integral part of the team that has developed award-winning projects like Highland bridge stormwater project. Municipal bodies are required to construct stormwater volume reduction Best Management Practices (BMPs) concurrent with projects that generate or reconstruct impervious surfaces. Each year, the BMP system my team designs for city of Saint Paul collects and treats millions of gallons of rainwater, keeping tons of sediment and harmful chemicals out of the Mississippi River.

Bush-Desoto Stormwater Infiltration Pond

City of Saint Paul, MN | 2024–2028 |

The Bush-Desoto Stormwater Infiltration Pond is a stormwater treatment and infiltration green infrastructure project located in Saint Paul, Minnesota. Provided technical review for developing cost effective and practical operation and maintenance strategy. Meeting the City's stormwater management goals and refining current design and maintenance practices to support broader sustainability, habitat restoration, and public-space goals.

Flandrau-Case Stormwater Infiltration Pond

City of Saint Paul, MN | 2024–2026 |

The Flandrau-Case Stormwater Infiltration Pond is a stormwater treatment and infiltration green infrastructure project located in Saint Paul, Minnesota. Provided technical review for developing cost effective and practical operation and maintenance strategy. Assisting the City's stormwater management goals and refining current design and maintenance practices.

Chester Street, Custer Street and Moses Street flood gate

Saint Paul MN | Technical reviewer

Technical review of Chester Street, Custer Street and Moses Street flood gate replacement projects. These projects are aimed at modernizing the historic Mississippi river flood protection structures.

II. Past Projects

Pump & Lift Stations

Kentucky American Water KRS1 Transfer Pumps Replacement

Kentucky American Water | Lexington, KY | ~USD 250K | Civil Engineer in Training

Ongoing project replacing an 18-MGD fixed pump with a variable-speed pump. Performed hydraulic analysis and design of the existing raw water system to inform transfer pump capacity decisions.

Ford-Hampton Booster Station

Kentucky American Water | Lexington, KY | Hydraulic Modeler

Ongoing upgrade of the drinking water system supplying southern Clark County along Boonesboro Road (KY State Route 627), including water main replacement and a new booster station replacing a 2021 emergency-supply booster station. Revised the hydraulic model and performed pump sizing to meet demand and head requirements.

Kentucky American Water Cox Street Booster Station Design

Lexington, KY | Hydraulic Modeler

Performed hydraulic analysis of existing and proposed conditions using WaterGEMS; ran scenario analysis across demand and flow conditions to confirm the proposed improvement met regulatory requirements.

Kentucky American Water KRS1 Low Services Pump

Kentucky American Water | Lexington, KY | 2024 | Hydraulic Designer

Hydraulic design of a new 15-MGD variable-frequency-drive pump for a 45-MGD raw water intake station, part of a long-term infrastructure improvement program at Kentucky River Station I. Developed the raw water main hydraulic model in WaterGEMS and authored front-end specifications for pump procurement.

Water Storage — Reservoirs & Tanks

Metro Water Bordeaux Hills Tank Replacement Project

Metro Water Services | Nashville, TN | Civil Designer

Project to replace two 500,000-gallon welded-steel storage tanks at the Bordeaux Hills Water Pump Station to increase storage in the City Low system and allow the new tank to "float" on that system. Performed hydraulic analysis for tank sizing and elevation to meet city requirements.

Aetna Mountain Tank

Tennessee American Water | Chattanooga, TN | Civil EIT

Ongoing project supporting finalization of water infrastructure for new developments, including a ~1-MG storage tank. Updated the hydraulic model to reflect the latest developer land-use plans and verified hydraulic performance and fire-flow requirements against current state guidelines.

Water Distribution & Transmission Pipelines

Quality Improvement Plan — Bell Court Area

Kentucky American Water | Lexington, KY | Civil Designer

Replacement of aging water mains at multiple locations in Lexington. Designed water mains and finalized replacement alignments, located hydrants and distribution mains, and developed the traffic control plan for construction permitting.

Millersburg Water Main Extension

Kentucky American Water | Hydraulic Modeler

Preliminary Planning Study to determine sizing and capacity of a water main connecting KAW's Central Division to the Millersburg system, sized for demand through 2050 and phased accordingly; current phase includes ~68,000 linear feet of water main replacing the existing Paris Water Works supply. Performed hydraulic modeling to confirm compliance with Kentucky Division of Water (KDOW) design criteria and conducted transient (surge) analysis on the transmission main.

Stormwater

Santee Cooper Winyah Generating Station

Georgetown, SC

Facility expansion project adding treatment capacity. Modeled existing and proposed storm sewer systems in PCSWMM to evaluate performance under additional runoff loading.

Santee Cooper Cross Generating Station

Pineville, SC

Facility expansion project. Modeled existing and proposed storm sewer systems, including replacement of storm mains and modification of the existing detention system, to evaluate performance of the designed storm system.

RESEARCH & TEACHING

Graduate Research Assistant

Aug 2021 – May 2023

University of Kentucky

Research centered on chlorine decay kinetics and disinfection byproduct (DBP) formation in drinking water distribution systems, including field sampling, regression modeling, and hydraulic model integration (KYPipe).

Instructional Roles

- Lab Instructor Assistant, Engineering Survey (CE 211) Fall 2022, Spring 2023
- Instructor, Computer Graphics and Communication / AutoCAD Lab (CE 106) Spring 2023

Current Research

- Prospective application of artificial intelligence in rural water infrastructure management for long-term sustainability.

PUBLICATIONS

- Unrine, J. M., McCoy, N., Christian, W. J., Gautam, Y., Ormsbee, L., & Sanderson, W. (2024). Spatial and seasonal variation in disinfection byproducts concentrations in a rural public drinking water system: A case study of Martin County, Kentucky, USA. *PLoS Water*, 3(3), e0000227. <https://doi.org/10.1371/journal.pwat.0000227>
- Fordjour, G., Gautam, Y., Ormsbee, L., Yost, S., & Unrine, J. (2025). Modeling the tradeoff between water loss, chlorine residuals, and trihalomethanes in rural Appalachia, USA. *Water*, 17(21), 3138. <https://doi.org/10.3390/w17213138>
- Gautam, Y. (2023). Prediction of chlorine and disinfection byproduct concentration in water distribution systems using KYPipe and TTHM regression models: Application to two systems in Kentucky [master's thesis, University of Kentucky]. UKnowledge. <https://doi.org/10.13023/etd.2023.241>

CONFERENCE PRESENTATIONS

- Gautam, Y., Gill, A., & Ormsbee, L. (2023, May). Development of model for TTHMs prediction: Application to rural water distribution systems in eastern Kentucky [Conference presentation]. EWRI Congress 2023, American Society of Civil Engineers.
- Unrine, J., Gautam, Y., McCoy, N., Mooney, M., Sexton, D., & Shrestha, S. (2024, November). Community-based participatory research methods to understand spatial and seasonal variation in drinking water disinfection byproducts in eastern Kentucky [Conference presentation]. SETAC North America 44th Annual Meeting.
- Lindell, O., Gill, A., & Gautam, Y. (2023, May). The development of digital twins for small rural water systems [Conference presentation]. EWRI Conference 2023, American Society of Civil Engineers.
- Gautam, Y. (2022, February). Modeling chlorine and disinfection by-products in water distribution systems in eastern Kentucky [Conference presentation]. 2022 Kentucky Water Resources Annual Symposium, Lexington, KY, United States.

TECHNICAL SKILLS

- Hydraulic Modeling Software: InfoWater Pro, WaterGEMS, InfoWorks ICM, KYPipe, PCSWMM, SewerGEMS, HAMMER, EPA SSOAP Toolbox
- Design & Drafting: AutoCAD, GIS
- Specialties: Water distribution & transmission design, wastewater/stormwater collection system modeling, pump & lift station design, water storage tank sizing, transient (surge) analysis, SWPPP development, chlorine decay & DBP modeling

ADDITIONAL RECOGNITIONS

- Graduate Assistantship, University of Kentucky (Aug 2021 – May 2023)
- Undergraduate Full Fellowship, Tribhuvan University (Oct 2011 – Apr 2015)